Boundary Conditions for Magnetic Fields

We now derive the boundary conditions satisfied by magnetic fields at the interface between two media.

1. Quick reminder from electrostatics

Before deriving the magnetic boundary conditions, recall the analogous electrostatic results:

- the **normal component of** $\vec{D}$ is discontinuous only if free surface charge exists,
- the **tangential component of** $\vec{E}$ is continuous in electrostatics.

That is,

$$\hat{n} \cdot (\vec{D}_2 - \vec{D}_1) = \rho_s, \qquad \hat{n} \times (\vec{E}_2 - \vec{E}_1) = 0. \tag{1}$$

Similarly, in magnetostatics we will derive the corresponding conditions for $\vec{B}$ and $\vec{H}$.

2. Boundary condition for the normal component of $\vec{B}$

We start from Gauss's law for magnetism:

$$\nabla \cdot \vec{B} = 0. \tag{2}$$

Integrating over a very thin pillbox straddling the boundary,

$$\iiint_V \nabla \cdot \vec{B}\, dv = \oiint_S \vec{B} \cdot d\vec{a} = 0. \tag{3}$$

As the height of the pillbox tends to zero, the flux through the side surface vanishes, and only the top and bottom faces contribute. Hence,

$$B_{2n}\, \Delta A - B_{1n}\, \Delta A = 0. \tag{4}$$

Therefore,

$$\boxed{B_{1n} = B_{2n}.} \tag{5}$$

So the **normal component of** $\vec{B}$ is always continuous across the boundary.

Result 1

$$\boxed{B_{1n} = B_{2n}}$$

The normal component of magnetic flux density is continuous across any interface.

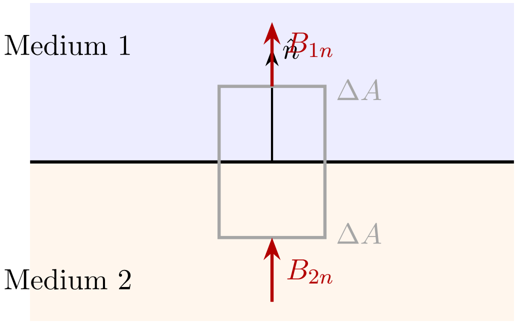

Figure 1: Pillbox argument at the interface of two media: the normal component of $\vec{B}$ is continuous.

3. Boundary condition for the tangential component of $\vec{H}$

Now start from Ampère's law in magnetostatics:

$$\nabla \times \vec{H} = \vec{J}_f. \tag{6}$$

Using Stokes' theorem on a very small rectangular loop crossing the boundary,

$$\oint_C \vec{H} \cdot d\vec{l} = I_{\text{enc,free}}. \tag{7}$$

Let the loop have length Δl parallel to the boundary and height Δh perpendicular to it. As $\Delta h \to 0$, the contributions from the short sides vanish, and we get

$$H_{2t}\Delta l - H_{1t}\Delta l = K_f \Delta l, \tag{8}$$

where K_f is the free surface current density at the interface.

Hence the general boundary condition is

$$\boxed{\hat{n} \times (\vec{H}_2 - \vec{H}_1) = \vec{K}_f.} \tag{9}$$

If there is **no free surface current** at the boundary, i.e. if

$$\vec{K}_f = 0, \tag{10}$$

then

$$\boxed{H_{1t} = H_{2t}.} \tag{11}$$

So the **tangential component of $\vec{H}$** is continuous only when no free surface current is present.

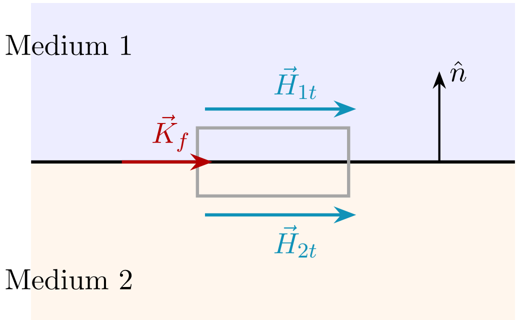

Figure 2: Amperian loop straddling the boundary: the jump in the tangential component of $\vec{H}$ is determined by the free surface current density $\vec{K}_f$.

Result 2

General form:

$$\boxed{\hat{n} \times (\vec{H}_2 - \vec{H}_1) = \vec{K}_f}$$

If there is no free surface current:

$$\boxed{H_{1t} = H_{2t}}$$

4. What about the tangential component of $\vec{B}$?

Since

$$\vec{B} = \mu \vec{H}, \tag{12}$$

the tangential component of $\vec{B}$ depends on the permeability of the media.

If there is no free surface current, then

$$H_{1t} = H_{2t}. \tag{13}$$

Therefore,

$$\frac{B_{1t}}{\mu_1} = \frac{B_{2t}}{\mu_2}. \tag{14}$$

Hence,

$$\boxed{B_{1t} = \frac{\mu_1}{\mu_2} B_{2t}.} \tag{15}$$

So the tangential component of $\vec{B}$ is **not generally continuous**. It is continuous only when the two media have the same permeability.

Important Remark

$$\boxed{B_{1n} = B_{2n}}$$

but

$$\boxed{B_{1t} \neq B_{2t} \text{ in general.}}$$

If $K_f = 0$, then

$$\boxed{\frac{B_{1t}}{\mu_1} = \frac{B_{2t}}{\mu_2}}$$

because the tangential component of $\vec{H}$ is continuous.

Boxed Summary

For magnetic fields at a boundary:

$$\boxed{B_{1n} = B_{2n}} \qquad \text{(normal component of } \vec{B} \text{ is continuous)}$$

$$\boxed{\hat{n} \times (\vec{H}_2 - \vec{H}_1) = \vec{K}_f} \qquad \text{(jump in tangential } \vec{H} \text{ is due to free surface current)}$$

If $\vec{K}_f = 0$, then

$$\boxed{H_{1t} = H_{2t}}$$

and therefore

$$\boxed{\frac{B_{1t}}{\mu_1} = \frac{B_{2t}}{\mu_2}}.$$

So, in general:

- $\vec{B}_n$ is continuous,
- $\vec{H}_t$ is continuous only if there is no free surface current,
- $\vec{B}_t$ is generally discontinuous when $\mu_1 \neq \mu_2$.

Inductance: an Introduction

Magnetic field of a solenoid

Consider a solenoid carrying current I. For a long solenoid with N turns over length ℓ, the turns per unit length are

$$n = \frac{N}{\ell}. \tag{16}$$

For an *ideal infinitely long* solenoid, the magnetic field inside is uniform and is given by

$$\boxed{\vec{B} = \mu n I\,\hat{z}} \tag{17}$$

where $\mu = \mu_0 \mu_r$ is the permeability of the medium inside the solenoid.

In air or free space,

$$\vec{B} = \mu_0 n I\,\hat{z}. \tag{18}$$

For a *finite* solenoid, this expression is only approximate:

- the field is nearly uniform in the middle,
- the field spreads out near the ends (fringing),
- and the outside field is not exactly zero.

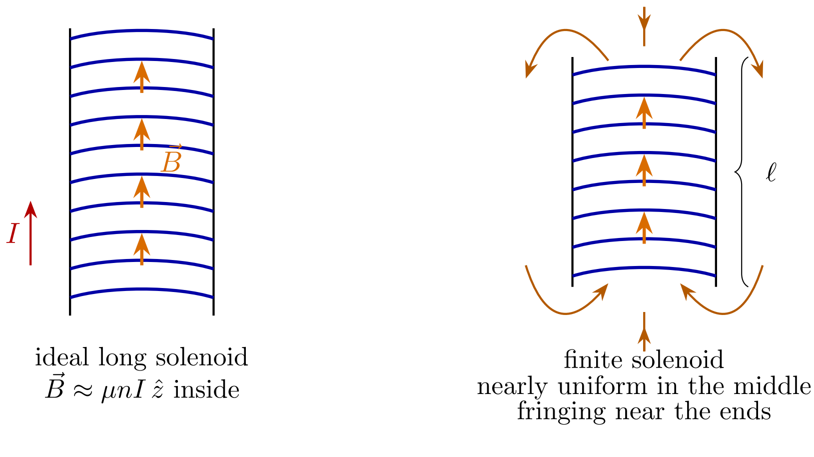

Figure 3: Magnetic field of a long solenoid and a finite solenoid. For a finite solenoid, the field spreads near the ends, so $\vec{B} \approx \mu n I\,\hat{z}$ is only an approximation.

Magnetic flux

Just as in electrostatics we defined electric flux through a surface, in magnetostatics we define the **magnetic flux** through a surface S as

$$\boxed{\Phi_m = \iint_S \vec{B} \cdot d\vec{s}.} \tag{19}$$

Here $d\vec{s}$ is the vector area element normal to the surface.

Definition

Magnetic flux is the total magnetic field passing through a chosen surface:

$$\Phi_m = \iint_S \vec{B} \cdot d\vec{s}.$$

Flux through different surfaces sharing the same boundary

Suppose two surfaces S_1 and S_2 have the same closed boundary C. Then the magnetic flux through them is the same.

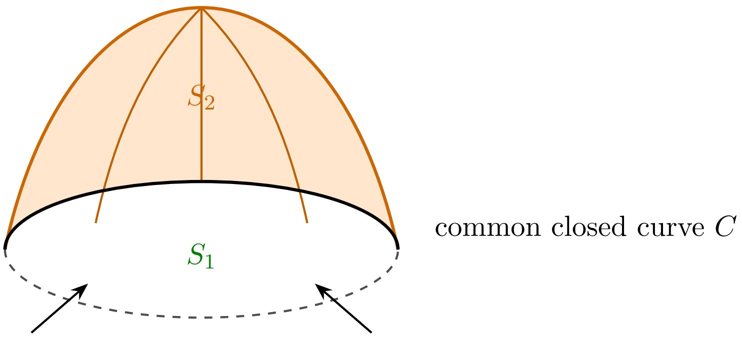

Figure 4: Two different surfaces S_1 and S_2 spanning the same closed boundary curve C.

To see this, consider the closed surface formed by $S_1 \cup (-S_2)$, where $(-S_2)$ means S_2 taken with opposite orientation. Then

$$\iint_{S_1} \vec{B} \cdot d\vec{s} + \iint_{-S_2} \vec{B} \cdot d\vec{s} = \oiint_{S_{\text{closed}}} \vec{B} \cdot d\vec{s}. \tag{20}$$

Using the divergence theorem,

$$\oiint_{S_{\text{closed}}} \vec{B} \cdot d\vec{s} = \iiint_V \nabla \cdot \vec{B}\, dv. \tag{21}$$

But

$$\nabla \cdot \vec{B} = 0. \tag{22}$$

Therefore,

$$\iiint_V \nabla \cdot \vec{B}\, dv = 0, \tag{23}$$

and hence

$$\boxed{\iint_{S_1} \vec{B} \cdot d\vec{s} = \iint_{S_2} \vec{B} \cdot d\vec{s}.} \tag{24}$$

So the magnetic flux depends only on the common boundary curve and not on which spanning surface is chosen.

Important consequence

If two surfaces share the same closed boundary, then the magnetic flux through them is the same:

$$\Phi_m(S_1) = \Phi_m(S_2).$$

This follows from $\nabla \cdot \vec{B} = 0$.

A useful comparison with electrostatics

In electrostatics, because

$$\nabla \times \vec{E} = 0,$$

the line integral of $\vec{E}$ between two points is path independent, and we define the electric potential difference:

$$V(\vec{r}_2) - V(\vec{r}_1) = -\int_{\vec{r}_1}^{\vec{r}_2} \vec{E} \cdot d\vec{l}. \tag{25}$$

In magnetostatics, because

$$\nabla \cdot \vec{B} = 0,$$

the magnetic flux through two surfaces spanning the same closed curve is independent of the chosen surface.

Careful remark

There is a *mathematical analogy* here:

$$\nabla \times \vec{E} = 0 \quad \Longrightarrow \quad \int \vec{E} \cdot d\vec{l} \text{ is path independent,}$$

$$\nabla \cdot \vec{B} = 0 \quad \Longrightarrow \quad \iint \vec{B} \cdot d\vec{s} \text{ is surface independent (for a fixed boundary).}$$

However, magnetic flux Φ_m is **not** the same physical quantity as electrostatic potential ϕ. The analogy is only in the mathematical structure.

Flux linkage and inductance

For a coil of N turns, if each turn is linked approximately by the same magnetic flux Φ_m, then the total **flux linkage** is

$$\boxed{\lambda = N\Phi_m.} \tag{26}$$

For a linear magnetic system, the flux linkage is proportional to the current:

$$\boxed{\lambda = LI.} \tag{27}$$

Here L is called the **inductance**. Thus,

$$\boxed{L = \frac{\lambda}{I}} \qquad \text{(linear case).} \tag{28}$$

For a single-turn loop, $\lambda = \Phi_m$, so

$$L = \frac{\Phi_m}{I}. \tag{29}$$

For a tightly wound N-turn solenoid,

$$L = \frac{N\Phi_m}{I}. \tag{30}$$

Definition of inductance

Inductance is the proportionality constant relating current and flux linkage:

$$\lambda = LI.$$

So inductance measures how much magnetic flux is linked by a circuit for a given current.

Comparison with the capacitor relation

In circuit theory, the capacitor relation

$$Q = CV \tag{31}$$

links stored charge with voltage.

Similarly, the inductor relation

$$\lambda = LI \tag{32}$$

links magnetic flux linkage with current.

This is why inductance becomes a basic circuit parameter, just as capacitance is.

Circuit-theory comparison

$$\boxed{Q = CV} \qquad \Longleftrightarrow \qquad \boxed{\lambda = LI}$$

- Q: charge stored in a capacitor
- V: voltage across the capacitor
- λ: flux linkage of an inductor
- I: current through the inductor

Boxed Summary

1. For a long solenoid,
$$\vec{B} \approx \mu n I \, \hat{z}.$$
For a finite solenoid, this is only approximate because of fringing near the ends.
2. Magnetic flux through a surface S is
$$\Phi_m = \iint_S \vec{B} \cdot d\vec{s}.$$
3. Because $\nabla \cdot \vec{B} = 0$, the magnetic flux through any two surfaces sharing the same closed boundary is the same.
4. For an N-turn coil, the total flux linkage is
$$\lambda = N\Phi_m.$$
5. For a linear magnetic system, inductance is defined by
$$\lambda = LI.$$
6. Thus inductance is the link between current and magnetic flux linkage, just as capacitance links voltage and charge.

Definition of Inductance

We now make the idea of inductance precise.

For any current-carrying circuit, let the magnetic flux through one turn be

$$\Phi_m = \iint_S \vec{B} \cdot d\vec{s}. \tag{33}$$

If the coil has N turns and each turn links approximately the same flux, then the total **flux linkage** is

$$\lambda = N\Phi_m. \tag{34}$$

The **inductance** is defined as the flux linkage per unit current:

$$\boxed{L = \frac{\lambda}{I} = \frac{N\Phi_m}{I}.} \tag{35}$$

For a single closed current loop, this reduces to

$$L = \frac{\Phi_m}{I}. \tag{36}$$

Definition

Inductance measures how much magnetic flux is linked with a circuit for a given current:

$$\boxed{L = \frac{\lambda}{I}}$$

where $\lambda = N\Phi_m$ is the total flux linkage.

1. Inductance of a long solenoid

Consider a long solenoid of length ℓ, cross-sectional area A, total number of turns N, carrying current I. Let

$$n = \frac{N}{\ell} \tag{37}$$

be the number of turns per unit length.

For a long solenoid, the magnetic field inside is approximately uniform:

$$\vec{B} \approx \mu n I \, \hat{z}. \tag{38}$$

Hence the magnetic flux through each turn is

$$\Phi_m = \iint \vec{B} \cdot d\vec{s} = BA = \mu n I A. \tag{39}$$

Therefore the total flux linkage is

$$\lambda = N\Phi_m = N(\mu n I A). \tag{40}$$

So the inductance is

$$L = \frac{\lambda}{I} = \frac{N(\mu n I A)}{I} = \mu n N A. \tag{41}$$

Since $N = n\ell$,

$$\boxed{L = \mu n^2 A \ell.} \tag{42}$$

Equivalently,

$$\boxed{L = \mu \frac{N^2 A}{\ell}.} \tag{43}$$

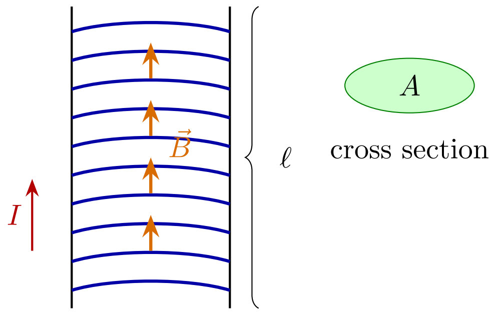

Figure 5: Long solenoid of length ℓ, cross-sectional area A, and N turns.

Solenoid result

$$\boxed{L = \mu n^2 A \ell = \mu \frac{N^2 A}{\ell}}$$

for a long solenoid.

2. Inductance of a toroid

Consider a toroidal coil of total turns N, inner radius a, outer radius b, and axial height h, carrying current I.

From Ampère's law, the magnetic field inside the toroid is

$$\vec{B}(r) = \mu \vec{H}(r) = \mu \frac{NI}{2\pi r}\,\hat{\phi}, \qquad a < r < b. \tag{44}$$

Take a differential strip of the cross-section of width dr and height h. Then

$$ds = h\,dr. \tag{45}$$

Hence the flux through one turn is

$$\Phi_m = \int_a^b B(r)\,h\,dr = \int_a^b \mu\frac{NI}{2\pi r}h\,dr. \tag{46}$$

So,

$$\Phi_m = \frac{\mu NIh}{2\pi}\int_a^b \frac{dr}{r} = \frac{\mu NIh}{2\pi}\ln\left(\frac{b}{a}\right). \tag{47}$$

Therefore the total flux linkage is

$$\lambda = N\Phi_m = N\cdot\frac{\mu NIh}{2\pi}\ln\left(\frac{b}{a}\right). \tag{48}$$

Hence the inductance is

$$\boxed{L = \frac{\lambda}{I} = \frac{\mu N^2 h}{2\pi}\ln\left(\frac{b}{a}\right).} \tag{49}$$

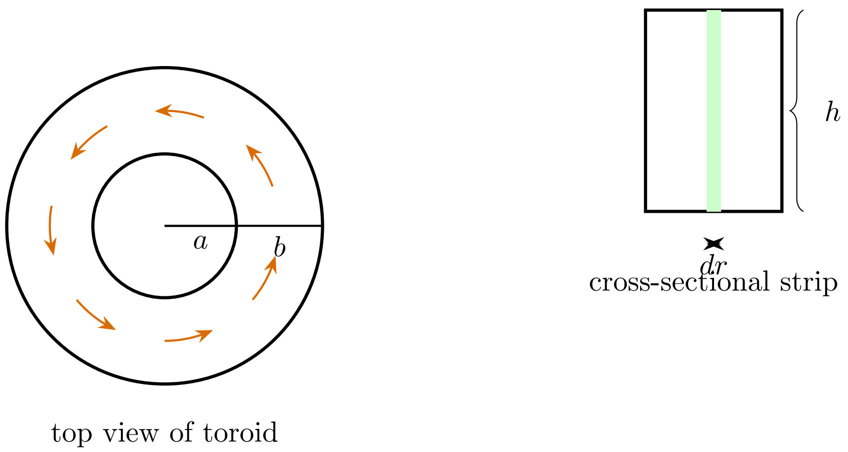

Figure 6: Toroid of inner radius a, outer radius b, and axial height h.

Toroid result

$$\boxed{L = \frac{\mu N^2 h}{2\pi}\ln\left(\frac{b}{a}\right)}$$

for a toroid of rectangular cross section.

3. Inductance of a coaxial cable

Consider a coaxial cable of length ℓ, inner conductor radius a, and inner radius of the outer conductor b. Assume current I flows in the inner conductor and returns through the outer conductor.

For the region $a < r < b$, Ampère's law gives

$$\vec{B}(r) = \mu \frac{I}{2\pi r} \hat{\phi}. \tag{50}$$

To find the external flux linkage, choose a radial strip of length ℓ and width dr. Then

$$ds = \ell \, dr. \tag{51}$$

Hence the magnetic flux linked with the line is

$$\Phi_m = \int_a^b B(r) \, \ell \, dr = \int_a^b \mu \frac{I}{2\pi r} \ell \, dr. \tag{52}$$

Therefore,

$$\Phi_m = \frac{\mu I \ell}{2\pi} \int_a^b \frac{dr}{r} = \frac{\mu I \ell}{2\pi} \ln\left(\frac{b}{a}\right). \tag{53}$$

Since this is a one-loop current path, the flux linkage is

$$\lambda = \Phi_m. \tag{54}$$

Therefore the inductance is

$$\boxed{L = \frac{\lambda}{I} = \frac{\mu \ell}{2\pi} \ln\left(\frac{b}{a}\right).} \tag{55}$$

So the inductance per unit length is

$$\boxed{L' = \frac{L}{\ell} = \frac{\mu}{2\pi} \ln\left(\frac{b}{a}\right).} \tag{56}$$

Remark: This is the **external inductance**, obtained from the magnetic field in the dielectric region $a < r < b$. Internal inductance of the conductors is neglected, which is a good approximation at high frequency.

Coaxial cable result

$$\boxed{L = \frac{\mu \ell}{2\pi} \ln\left(\frac{b}{a}\right)} \qquad \boxed{L' = \frac{\mu}{2\pi} \ln\left(\frac{b}{a}\right)}$$

for the external inductance of a coaxial cable.

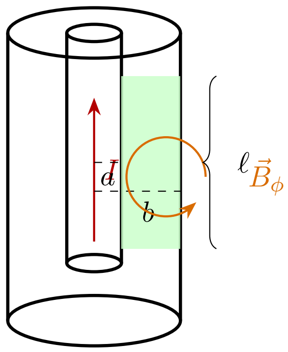

Figure 7: Coaxial cable of length ℓ, inner radius a, and outer-conductor inner radius b.

Final comparison of the three cases

$$\boxed{L_{\text{solenoid}} = \mu \frac{N^2 A}{\ell}}$$

$$\boxed{L_{\text{toroid}} = \frac{\mu N^2 h}{2\pi} \ln\left(\frac{b}{a}\right)}$$

$$\boxed{L_{\text{coax}} = \frac{\mu \ell}{2\pi} \ln\left(\frac{b}{a}\right)}$$

These results follow directly from:

$$L = \frac{\lambda}{I}, \qquad \lambda = N\Phi_m, \qquad \Phi_m = \iint_S \vec{B} \cdot d\vec{s}.$$

Mutual Inductance

We now extend the idea of self inductance to the case of **two nearby current-carrying loops**. Because magnetic fields obey superposition, the field produced by one loop can link the other loop and vice versa. This gives rise to **mutual inductance**.

1. Two coupled loops

For clarity, let us first consider two *single-turn* loops:

- loop 1 carries current I_1,

- loop 2 carries current I_2.

Let the magnetic fields produced by these currents be $\vec{B}_1$ and $\vec{B}_2$, respectively. Then the total magnetic field is

$$\vec{B} = \vec{B}_1 + \vec{B}_2. \tag{57}$$

Figure 8: Two nearby current-carrying loops. The field of loop 1 links loop 2 and the field of loop 2 links loop 1. The right sketch shows the geometry used in the Neumann formula.

2. Flux through each loop

The total flux through loop 2 is

$$\Phi_{m2} = \iint_{S_2} \vec{B} \cdot d\vec{s} = \iint_{S_2} (\vec{B}_1 + \vec{B}_2) \cdot d\vec{s}. \tag{58}$$

Hence

$$\Phi_{m2} = \iint_{S_2} \vec{B}_1 \cdot d\vec{s} + \iint_{S_2} \vec{B}_2 \cdot d\vec{s}. \tag{59}$$

Now:

- the second term is the **self-flux** of loop 2 due to I_2,
- the first term is the **mutual flux** in loop 2 due to current I_1.

So we write

$$\boxed{\Phi_{m2} = M_{21} I_1 + L_2 I_2.} \tag{60}$$

Similarly, the total flux through loop 1 is

$$\boxed{\Phi_{m1} = L_1 I_1 + M_{12} I_2.} \tag{61}$$

Here:

- L_1 and L_2 are the self inductances,
- M_{12} and M_{21} are the mutual inductances.

Meaning of mutual inductance

$$M_{21} = \frac{\text{flux through loop 2 due to current in loop 1}}{I_1}, \qquad M_{12} = \frac{\text{flux through loop 1 due to current in loop 2}}{I_2}.$$

3. Matrix form for two loops

Equations (61) and (60) may be written in matrix form as

$$\boxed{\begin{bmatrix} \Phi_{m1} \\ \Phi_{m2} \end{bmatrix} = \begin{bmatrix} L_1 & M_{12} \\ M_{21} & L_2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}.} \tag{62}$$

The diagonal elements are self inductances, and the off-diagonal elements are mutual inductances.

For N coupled loops, the same idea generalizes to

$$\boxed{\begin{bmatrix} \Phi_{m1} \\ \Phi_{m2} \\ \vdots \\ \Phi_{mN} \end{bmatrix} = \begin{bmatrix} L_1 & M_{12} & \cdots & M_{1N} \\ M_{21} & L_2 & \cdots & M_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ M_{N1} & M_{N2} & \cdots & L_N \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_N \end{bmatrix}.} \tag{63}$$

Inductance matrix

For many coupled loops, flux and current are related by

$$\boxed{\boldsymbol{\Phi}_m = \mathbf{M}\,\mathbf{I}}$$

where the diagonal elements of $\mathbf{M}$ are self inductances and the off-diagonal elements are mutual inductances.

4. Mutual inductance using the magnetic vector potential

Now let us derive a useful formula for mutual inductance.

The mutual flux in loop 2 due to current in loop 1 is

$$\Phi_{21} = \iint_{S_2} \vec{B}_1 \cdot d\vec{s}. \tag{64}$$

Since

$$\vec{B}_1 = \nabla \times \vec{A}_1, \tag{65}$$

we get

$$\Phi_{21} = \iint_{S_2} (\nabla \times \vec{A}_1) \cdot d\vec{s}. \tag{66}$$

By Stokes' theorem,

$$\Phi_{21} = \oint_{C_2} \vec{A}_1 \cdot d\vec{\ell}_2. \tag{67}$$

Now the vector potential due to loop 1 carrying current I_1 is

$$\vec{A}_1(\vec{r}_2) = \frac{\mu_0 I_1}{4\pi} \oint_{C_1} \frac{d\vec{\ell}_1}{R_{12}}, \tag{68}$$

where

$$R_{12} = |\vec{r}_2 - \vec{r}_1|. \tag{69}$$

Substituting into Eq. (67),

$$\Phi_{21} = \frac{\mu_0 I_1}{4\pi} \oint_{C_2} \oint_{C_1} \frac{d\vec{\ell}_1 \cdot d\vec{\ell}_2}{R_{12}}. \tag{70}$$

Therefore,

$$\boxed{M_{21} = \frac{\Phi_{21}}{I_1} = \frac{\mu_0}{4\pi} \oint_{C_2} \oint_{C_1} \frac{d\vec{\ell}_1 \cdot d\vec{\ell}_2}{R_{12}}.} \tag{71}$$

This is called the **Neumann formula**.

5. Important consequences of Neumann's formula

The Neumann formula immediately gives two important insights.

(i) Mutual inductance is a geometrical quantity

For fixed medium permeability, the quantity

$$M_{21} = \frac{\mu_0}{4\pi} \oint_{C_2} \oint_{C_1} \frac{d\vec{\ell}_1 \cdot d\vec{\ell}_2}{R_{12}}$$

depends only on:

- the shapes of the two loops,

- their sizes,
- and their relative positions and orientations.

So mutual inductance is fundamentally a **geometrical coupling parameter**.

(ii) Reciprocity: $M_{12} = M_{21}$

Because the double integral is unchanged when we interchange loop 1 and loop 2, we get

$$\boxed{M_{12} = M_{21}.} \tag{72}$$

So the mutual inductance matrix is symmetric for ordinary reciprocal media.

Neumann formula and reciprocity

$$\boxed{M_{21} = \frac{\mu_0}{4\pi} \oint_{C_2} \oint_{C_1} \frac{d\vec{\ell}_1 \cdot d\vec{\ell}_2}{R_{12}}}$$

shows that:

1. mutual inductance depends only on geometry,
2. and

$$\boxed{M_{12} = M_{21}}.$$

6. Remark on sign and orientation

The sign of the mutual term depends on the chosen reference directions for current and flux linkage. With consistent loop orientations:

- $M > 0$ for *aiding* coupling,
- $M < 0$ for *opposing* coupling.

In circuit theory, this is handled using the **dot convention**. In many geometry-only derivations, we focus first on the magnitude and reciprocity.

Boxed Summary

For two coupled loops:

$$\boxed{\Phi_{m1} = L_1 I_1 + M_{12} I_2, \qquad \Phi_{m2} = M_{21} I_1 + L_2 I_2.}$$

In matrix form:

$$\boxed{\begin{bmatrix} \Phi_{m1} \\ \Phi_{m2} \end{bmatrix} = \begin{bmatrix} L_1 & M_{12} \\ M_{21} & L_2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}}$$

And the mutual inductance is given by the Neumann formula:

$$\boxed{M_{21} = \frac{\mu_0}{4\pi} \oint_{C_2} \oint_{C_1} \frac{d\vec{\ell}_1 \cdot d\vec{\ell}_2}{R_{12}}}$$

This immediately shows that:

$$\boxed{M_{12} = M_{21}}$$

and that mutual inductance is determined by geometry.